Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/08/2025
PI Name:	Strangman, Gary E Ph.D.		
Project Title:	Quantifying and Predicting Operationally-Relevant Performance in a Long-Duration Spaceflight Analog		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sleep**:Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	strang@nmr.mgh.harvard.edu	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	617-724-0662
Organization Name:	Massachusetts General Hospital		
PI Address 1:	Department of Psychiatry		
PI Address 2:	149 13th Street, Suite 2651		
PI Web Page:			
City:	Charlestown	**State:**	MA
Zip Code:	02129-2020	**Congressional District:**	7
Comments:			
Project Type:		**Solicitation / Funding Source:**	03-OBPR-02
Start Date:	08/01/2016	**End Date:**	04/30/2025
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 4/30/2025 per NSSC information (Ed., 5/2/24) NOTE: End date changed to 4/30/2024 per V. Lehman/JSC (Ed., 5/16/23) NOTE: End date changed to 4/30/2023 per NSSC information (Ed., 5/2/22) NOTE: End date changed to 4/30/2022 per NSSC information (Ed., 4/12/21) NOTE: End date changed to 4/30/2021 per NSSC information (Ed., 5/4/2020) NOTE: Changed end date to 9/30/2020 per L. Juliette/HRP (Ed., 2/19/2020) NOTE: Extended to 1/31/2020 per K. Ohnesorge/HRP JSC (Ed., 5/24/18) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/18/17)		
Key Personnel Changes/Previous PI:			

COI Name (Institution):	Zhang, Quan Ph.D. (Massachusetts General Hospital) Ivkovic, Vladimir Ph.D. (Massachusetts General Hospital)
Grant/Contract No.:	NNX16AO30G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Exploration spaceflight missions will expose crewmembers to many risks that could affect their performance and mission success. Minimizing such risks will require identifying and validating objective indicators of behavioral health and performance (BMed-102 Gap), understanding the contribution of sleep loss on individual behavioral health (Sleep-101 Gap), and identifying countermeasures that can reduce these risks (BMed-103 Gap). Currently the Robotic On-Board Trainer (ROBoT) is used operationally by astronauts both on the ground and on the International Space Station (ISS) to practice Canada Arm activities. Our group developed ROBoT for research use (ROBoT-r) and for quantitative performance assessment. In addition, our group is developing and testing NINscan-SE: a multi-use system for measuring brain and physiological function. Both ROBoT-r and NINscan-SE were deployed during the Human Exploration Research Analog (HERA) Campaigns 4 and 5. We will deployed both systems to: Aim 1: Characterize operational task performance changes during 45-day HERA missions, including the roles of time-in-mission, workload, sleep debt, and operational emergencies. Aim 2: Characterize brain and systemic physiology changes during 45-day HERA missions, including the roles of time-in-mission, workload, sleep debt, and operational emergencies. Aim 3: Identify physiological or behavioral variables that predict operational performance. Aim 4: Quantify the influence of behavioral health countermeasures on both operational performance and (neuro)physiological measures. To achieve these aims, we recruited 36 crewmembers from nine 45-day missions in the HERA facility during Campaigns 4 and 5, plus 24 control subjects. HERA and control participants will all perform ROBoT tasks plus undergo physiological monitoring 2-3x/week, on matching schedules, thus enabling us to differentiate changes in operational performance due to practice over time from any changes due to HERA sequestration. In addition, two “unexpected operational emergency” events will be introduced in the first and last weeks of each HERA mission. These will consist of an acute need to capture a wayward satellite traveling near the limits of Canada Arm capabilities. We also worked with the Human Factors and Behavioral Performance (HFBP) Element and other HERA investigators to coordinate ROBoT and physiological data collection before, during, and after one or more countermeasure (CM) deployments during the HERA missions. A sleep restriction intervention was included in HERA C4. In addition, CMs included a lighting intervention, a diet intervention, and a habitat (volume and privacy) intervention. We tested hypotheses that the CM(s) generate detectable changes in ROBoT performance and rest/task (neuro)physiology recordings. We also compared ROBoT performance to the standardized Cognition Test Battery. The knowledge-deliverables of this project include: (i) changes in operationally-relevant (ROBoT-r) performance during the HERA mission in a well-controlled analog study of substantial size; (ii) changes in cerebral and systemic physiology associated with HERA mission parameters as well as operational performance; (iii) identification of potential predictors of future ROBoT-r performance; and (iv) the influence of the investigated countermeasure(s) on operational performance and physiology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ROBoT system—and the HERA isolation protocol—are quite specific to NASA spaceflight operations and hence have relatively few direct Earth applications. However, the ROBoT spacecraft-capture simulations represent a highly skilled, complex, operational performance task. It could thus be used as a comparison task in concert with detailed cognitive testing to help dissect the cognitive components complex tasks as well as the influence of other physiological stressors (e.g., sleep deprivation, alcohol consumption, medical radiation) on the performance of such tasks. Use of different complex tasks with the same approach could be useful in assessing and predicting performance in a wide range of other operational environments (diving, pilots, military, surgeons, etc.). Regarding NINscan-SE, no current NIRS (near-infrared spectroscopy), EEG, or polysomnography device has both the portability and the multi-use features of the system we deployed. This system could thus have substantial novel Earth applications. Hospital monitoring applications could include long-duration, non-invasive brain monitoring in the NeuroICU following stroke or traumatic injury, for which no similar technology exists. Real-time, in-office brain activation assessment could also be supported, for assessment of psychiatric states, for monitoring the neural effects of cardiovascular or psychoactive drugs or other therapies, or for brain monitoring during rehabilitation. Mobile monitoring could perhaps have an even larger impact outside the hospital setting. A wearable monitor would enable ambulatory syncope monitoring, or multi-parameter ambulatory epilepsy monitoring. If deployed in emergency settings, NINscan-SE could potentially be used to detect cerebral or abdominal hemorrhage, ischemia, and/or cortical spreading depression by first responders. Home monitoring uses include various sleep disorders, as well as various commercial possibilities.
	The overall goal of this project is to assess operationally-relevant behavioral performance over 45-day isolation and confinement periods in the Human Exploration Research Analog (HERA), as well as associated neurophysiological status during HERA missions. Operational performance was evaluated using the ROBoT-r task—an operationally used track-and-capture task for grappling incoming resupply vehicles using Canadarm2. The ROBoT-r task was modified for research use as part of the separate Behavioral Core Measures project by Drs. Strangman and Ivkovic, first deployed in HERA Campaign 3. Brain and systemic physiological assessments include resting-state heart rate (HR), heart rate variability (HRV), cerebrovascular pulsatility, and EEG, plus functional brain activation during the ROBoT-r task trials using our near-infrared spectroscopy and imaging (NIRS/NIRI) based NINscan devices, as well as HR, HRV, pulsatility, EOG, and EEG during ROBoT-r performance. In addition, this work included a supplement

Task Progress:

project—BRAIN-STIM—that sought to test the ability of transcranial direct current stimulation (tDCS) to enhance ROBoT-r performance.
Summary and Takeaways

While we strongly encourage the reader to review the entire report to understand our results and statements in proper context, here we list some key points from across this project that we consider particularly important to highlight.

Aim 1: Characterize operationally-relevant task performance changes during 45-day HERA missions. We thoroughly characterized operationally-relevant task performance changes associated with 45-day HERA missions across numerous behavioral metrics. With regard to our specific hypotheses: • Hypothesis 1.1: Overall ROBoT-r performance in HERA crewmembers will differ significantly from controls, and will be related to time-in-mission. o We found that being in HERA sequestration led to poorer performance on ROBoT-r, and that there was a significant improvement in performance over time-in-mission, driven by a strong learning curve that lasted through the majority of the HERA sequestration period. • Hypothesis 1.3: Overall ROBoT-r performance will be significantly poorer during emergency procedures than during ordinary ROBoT performance. o Contrary to our hypothesis, when comparing apples-to-apples (i.e., 100% difficulty trials, which were the only type of trials conducted on emergency days), we unexpectedly found performance during emergency procedures to be equivalent to behavioral performance on adjacent days. So, despite the high difficulty and the unscheduled nature of emergency trials, participants were able to handle those without performance reductions. We, therefore, ignored these sessions in following analyses.

Aim 2: Characterize brain and systemic physiology changes during 45-day HERA missions. We also thoroughly characterized a number of systemic physiology and brain changes during the 45-day HERA missions. With regard to our specific hypotheses: • Hypothesis 2.1: NIRS measures of prefrontal cortical activity during the ROBoT task will vary as a function of time-in-mission and sleep debt. o Time-in-Mission Resting HRV increased; whereas, during ROBoT-r HRV increased and the following all decreased: HRslope, Pcardiac, Presp, and L-DLPFC, Fp1, and Fp2 activation levels. o Sleep Restriction HR during ROBoT-r tasks was elevated with sleep restriction

Aim 3: Identify physiological or behavioral variables that predict operational performance. Since the Cognition Test Battery (CTB) and ROBoT-r were often conducted on the same day, we used Cognition outcome metrics to predict ROBoT-r performance. We identified four tests within the CTB whose outcome metrics provided predictive value for ROBoT-r weighted_scores: Abstract Matching, Motor Praxis Test, Emotion Recognition, and the Line Orientation Test. • Hypothesis 3.1: One or more physiological variables will exhibit significant relationships with overall performance on ROBoT-r trials. o We investigated whether EEG power spectra in the rest period could predict subsequent ROBoT-r performance, but found no significant relationships. We did find significant relationships with ROBoT-r (in the theta and alpha bands) during task performance, although those relationships would not be useful for predicting future performance.

Aim 4: Quantify the influence of behavioral health countermeasures on both operational performance and (neuro)physiological measures. While we did not implement any countermeasures of our own in this study, HERA C4/C5 included one major intervention (Sleep Restriction) and three separate countermeasures (Lighting, Diet, and Habitat Size). As discussed, we were unable to investigate Habitat Size. However, we were able to test for the effects of Lighting and Diet interventions. The experimental Lighting condition improved ROBoT-r performance (weighted_score and overall_success, with no effect on duration), largely counteracting the detrimental effects of Sleep Restriction. Diet did not significantly affect ROBoT-r performance overall, which is encouraging.

BRAIN-STIM Aim: Test whether transcranial electrical stimulation can improve ROBoT-r performance, as a potential spaceflight countermeasure. We conducted a thoroughly double-blind, sham-controlled study of brain stimulation over the L-DLPFC and the L-aINS in n=40 participants. • Hypothesis S1: tDCS will enhance performance on ROBoT-r compared to sham stimulation. o Contrary to our hypothesis, tDCS did not enhance performance and, in fact, had a small but statistically significant negative influence on ROBoT-r performance variables. While this was an unexpected result, and various explanations remain to be tested, we suspect the reason lies partly in the complex nature of the ROBoT-r task—which engages many brain regions simultaneously—and the rather large brain regions stimulated by tDCS.

Additional Key Details Here, we list some key details from across this project that we consider important to highlight.

ROBoT-R: • Isolation: There was a significant decrease in performance (poorer accuracy and lower probability of success) when participants were sequestered in HERA versus when they were outside of HERA (pre-/post-mission). • Lighting: The experimental lighting condition in HERA C4 significantly increased accuracy (Weighted Score), did not change speed (Duration), and significantly increased (near doubling) of the probability of overall success compared to control lighting. This highlights the role of sleep in behavioral performance and suggests that the dynamic lighting countermeasure (implemented by Dr. Steven Lockley) was a highly effective modulator of ROBoT-r performance. • Early Performance Disruptions: During the first two weeks after ICE ingress, there appeared to be a slowed learning process, with less improvement in accuracy or overall success and no improvement in speed compared to Controls. This could be related to the elevated stress, workload, sleep restriction, or more general “orientation” required upon entering a dramatically new environment. This is consistent with spaceflight reports of impaired performance and added workload/distraction in the days before and after both launch and landing. • ROBoT-r Control: Users of ROBoT seem to emphasize correcting translational error over rotational error, suggesting a different strategic control approach for the two hands. It also suggests that the lesser-attended modality—rotation—is likely to provide a more sensitive indicator of performance status. • Reversals and Collisions: Learning to avoid reversals and collisions seemed to occur more rapidly than for other metrics, plateauing after ~10 sessions as compared to the plateau after ~17 sessions for other variables. • C5 Scoring: The change in scoring of ROBoT-r between HERA C4 and C5 means that the training process for C5 (using the identical video and trainer guidance) was, in fact, inaccurate. Participants were told that reversals mattered and efficiency (=run duration) did not matter; whereas, the opposite was the case throughout C5. Moreover, merging efficiency(=duration) into weighted_score as of 2021 is particularly concerning, given that this fails to maintain a separation of accuracy from speed measures. As we understand it, this problem may still persist and is basically invisible to nearly all researchers. Our approach was to recompute weighted_score so that it was computed consistently throughout this study (2017-2024).

Physiology: • Brain Activation: In L-DLPFC, Fp2, and R-DLPFC exhibited significantly lower [HbD] changes, while in HERA, suggesting lower brain activation in this region while sequestered. This effect was almost completely compensated for by the Lighting intervention (both effect sizes ~5 M). • Cerebrovascular Pulsatility: While the meaning of cerebrovascular pulsatility changes is not yet fully understood, we did identify notably consistent responses across the different types of pulsatility: decreases with Mission Day, increases with Difficulty, and decreases in HERA,

	with sleep restriction, and with the Lighting and Diet countermeasures. This consistency merits further investigation. BRAIN-STIM: Complex Tasks: The simple application of tES during a complex task such as ROBoT-r may not produce effects similar to those observed with simpler tasks. Given the distributed nature of tES stimulation in the brain, it will be important to conduct studies with the specific tasks to confirm and quantify relevant behavioral effects.
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